Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Crescendo
Project / Solution Name	OpusEvents
Name (Leader)	Yaw En Qi
Email & Phone (Leader)	shaniceyaw@gmail.com, +60 162682917
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://opusevent-event-planner.replit.app/opusevents/sign-in
Code Repository URL (public / access granted to judges)	https://github.com/shaniceyaw-creator/SDC_Hacakathon_Crescendo

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Orbyte EventOps connects frontend user interactions to background AI modeling, cloud APIs, and persistent database storage.

A React 19 and TypeScript frontend provides dedicated Manager, Worker, and Customer role views. Generated OpenAPI clients route requests to an Express 5 API server that validates input with Zod and verifies signed demo role cookies. Express sends JSON commands to a persistent Python backend, where self-trained scikit-learn regression models predict required crew, equipment, total cost, and selling price.

Deterministic business rules cross-reference staff availability against Orbyte eLeave records, storing bookings, tasks, expenses, and audit logs in PostgreSQL. An OpenAI manager assistant streams plain-English advisory text via Replit AI Integrations. Receipt attachments are securely stored in Replit App Storage with Replit OpenID Connect identity verification. The complete system is deployed live on Replit Core.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.13.11 workspace / 3.12.3 model training	Core planning engine, numerical calculations, and local model inference	PSF License
TypeScript 5.9 / Node.js 24	Typed frontend components and Express API backend	Apache-2.0 / MIT
React / React DOM 19.1	Component-based user interface and role views	MIT
Vite 7.3 / esbuild 0.28	Development bundling and production build toolchain	MIT
Tailwind CSS 4.3 / Radix UI	Responsive styling and accessible UI primitives	MIT
TanStack React Query 5.102	Asynchronous state management and client-side caching	MIT
Zod 3.25 / React Hook Form 7.87	Schema validation and form handling	MIT
Express 5.2 / cookie-parser	HTTP API routing, signed cookies, and middleware handling	MIT
pandas 2.3 / NumPy 2.5	Data processing, CSV parsing, and matrix operations	BSD-3-Clause
scikit-learn 1.9 / joblib 1.6	Multi-target regression modeling and model serialization	BSD-3-Clause
psycopg 3.3 / PostgreSQL	Database driver and persistent PostgreSQL relational storage	LGPL-3.0 / PostgreSQL License
Drizzle ORM 0.45 / pg 8.23	ORM toolkit for receipt session handling and schema migration	Apache-2.0 / MIT

Name & Version	Purpose in Project	License
Leaflet 1.9 / React Leaflet 5.0	Interactive venue map rendering	BSD-2-Clause / Hippocratic-2.1
OpenAI JS SDK 6.49	Integration client for OpenAI manager assistant	Apache-2.0
Leaflet 1.9 / React Leaflet 5.0	Interactive venue map rendering for event venue selection.	BSD-2-Clause / Hippocratic-2.1
Orval 8.30 / OpenAPI	Typed API client generation from OpenAPI specifications.	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenAI via Replit AI Integrations	Generates plain-English managerial summaries and operational advice.	Replit Managed Billing
Open-Meteo Forecast & Air Quality API	Supplies weather and environmental tips for outdoor event logistics	Public Endpoint (Free Tier)
OpenStreetMap Tile Service	Delivers map tiles for event venue mapping	Public ODbL Tile Service
Replit OpenID Connect (OIDC)	Identity verification for private receipt attachment access	Replit Managed Identity
Replit App Storage (GCS SDK)	Private cloud storage for uploaded expense receipts	Replit Managed Storage
Google Calendar Link Export	User-driven export of event schedules to ICS / web calendar.	Client-side export (No API Key)
Google Fonts CSS/font delivery	DM Sans/Manrope for the website; DM Sans/JetBrains Mono/Space Grotesk for the film.	Public font delivery; fonts under OFL-1.1.

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Publishing	Live public web hosting for frontend and Express/Python backend (https://opusevent-event-planner.replit.app/eventops-live/sign-in) .	Replit Autoscale
Replit PostgreSQL Database	Managed relational database storing bookings, schedules, tasks, and audit logs	Replit Managed Database
RunPod GPU Cloud	Offline cloud GPU server used to train the two-stage scikit-learn model bundle	Sponsor-Provided GPU Credits
GitHub	Version control repository hosting source code, schemas, and documentation	Public Repository
Replit App Storage	Private receipt bytes; bookkeeping metadata in PostgreSQL. Access mediated by authenticated API routes.	Managed storage via Google Cloud Storage SDK; tier/region/retention settings to confirm.
Local model/static assets	Saved model, synthetic CSV seeds, logos and film source/audio ship with the application.	Original uploaded archives removed; current model/training evidence retained. Local downloadable MP4 is not part of the website image.
Replit AI Integrations	Managed OpenAI API connectivity; application secrets are server-side.	Usage-billed / credits: team to confirm. No credentials included here.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
EventOps_v2 Resource Regressors	Self-trained (RandomForestRegressor, DecisionTreeRegressor, LinearRegression)	Self-trained offline on RunPod	BSD-3-Clause (scikit-learn)
EventOps_v2 Cost & Price Regressors	Self-trained two-stage LinearRegression model bundle	Self-trained offline on RunPod	BSD-3-Clause (scikit-learn)
gpt-5.6-terra (OpenAI)	Called via Replit AI Integrations	Pre-trained hosted language model	OpenAI / Replit Terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
events_history.csv (300 rows)	Created by Team (historical event logistics 2023–2026)	Synthetic	Public / Open Source	N
staff.csv & leave.csv	Created by Team (16 staff profiles, 18 leave entries)	Synthetic	Public / Open Source	N
upcoming_events.csv (74 rows)	Created by Team (initial database seed events)	Synthetic	Public / Open Source	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Open-Meteo & OpenStreetMap	Public API feeds (open-meteo.com, openstreetmap.org)	Public Data	CC BY 4.0 / ODbL 1.0	N
venue_areas.csv & festivals.csv	Reference data with Malaysian calendar context	Synthetic	Public / Team Synthetic	N
Original prototype CSVs retained for reference	data/malaysia_events_2025.csv, seed_events.csv, staff_assignments.csv, staff.csv, staff_leave.csv, event_rules.csv, inventory.csv	User-supplied synthetic / legacy data; not the active v2 model's evaluation set	Original provenance/licence requires team confirmation.	N
Weather / air-quality responses	https://open-meteo.com/; live external requests	Public provider data; not used to train the event model	Open-Meteo data attribution (CC BY 4.0) and underlying-provider/service terms apply.	N
User-created bookings, tasks, expenses and activity	Application forms → PostgreSQL	Runtime demo entries; synthetic values only in the public prototype	Team/customer permission and retention policy required for real deployment.	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent	Assistance with React UI layouts, Express API routing, and Python bridge integration.	Major
Anthropic Claude 3.5 Sonnet / ChatGPT	Code debugging, refactoring Python functions, and drafting documentation .	Moderate
AI Music & Image Generators	Creating promotional soundtrack audio and 9 event-type cover images.	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team thoroughly reviewed and tested every AI-assisted code function and text summary. We compiled TypeScript without errors, verified Python model predictions against manual spreadsheet calculations, and ensured all environment variables remain securely stored in Replit Secrets. External API outputs were verified for factual accuracy against our synthetic event database. The team leader retains full responsibility for all project deliverables.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
RunPod	Cloud GPU Credits	Provisioned a cloud GPU instance to train and evaluate the two-stage scikit-learn regression models offline
Replit	Replit Core & Publishing Services	Hosted the live web prototype, PostgreSQL database, App Storage, and Replit AI Integrations
Orbyte	Intended ecosystem, workflow context and displayed logo/technology credit.	Event-management module concept; future finance/HR handoff is described only. No live Orbyte API, data transfer or finance-role screen is implemented; confirm any actual organiser-provided resource.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

RunPod GPU credits enabled fast offline model training and parameter optimization for our 300-event dataset. Replit Core provided an integrated development and deployment environment for our live React and Express application. While local CPU training and alternative cloud platforms exist, sponsor resources significantly accelerated development velocity and ensured a stable, production-ready cloud deployment.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

All pre-seeded company, staff, and event records are 100% synthetic. Form submissions and PostgreSQL store event parameters, worker assignments, expenses, and audit entries.

Uploaded receipt files are stored as private objects in Replit App Storage, with access mediated by signed HttpOnly cookies and Replit OpenID Connect (OIDC) identity checks. Manager prompts sent to the OpenAI API contain event parameters and staff availability counts, strictly adhering to principles set under Malaysia's Personal Data Protection Act 2010 (PDPA).

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Primary technical risks include prediction variance for rare event types and potential GenAI API latency.

To mitigate these risks, OpusEvents implements deterministic business rules that enforce minimum equipment ratios and profit margin guardrails. External AI summaries are strictly advisory and cannot write directly to the database. If the external AI API experiences downtime or latency, local deterministic rules calculate baseline inventory and workforce requirements, ensuring 100% continuous operational uptime.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[24/10/2026]

Appendix A — System Block Diagram

Figure 9.1 shows the complete system block diagram

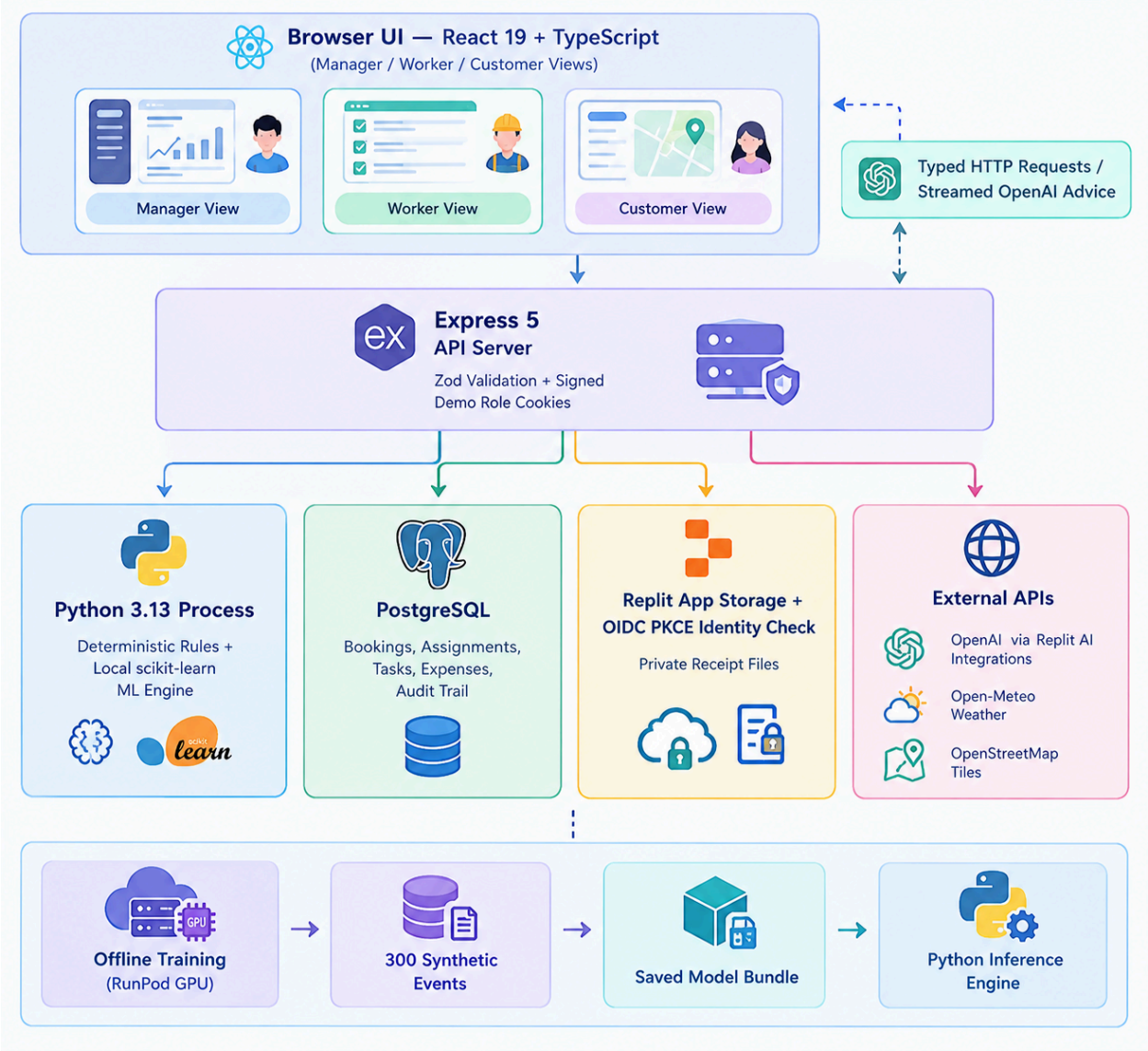


Figure 9.1 System Block Diagram